

An Analysis of the Law of Harmonies

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May 7, 2026

Introduction

Johannes Kepler published *Harmonices Mundi* (*The Harmony of the World*) in 1619, wherein he described his three laws of planetary motion. The third of these laws, now known as Kepler's Third Law or the Law of Harmonies, states that the square of a planet's orbital period is proportional to the cube of the semi-major axis of its orbit. That is,

$$P^2 \propto a^3,$$

where P is the orbital period of a given planet and a is the length of its semi-major axis. Kepler derived this relationship empirically, by hand, from the data of Tycho Brahe, whose observatory Kepler worked in.

Sixty-eight year later, in 1687, Isaac Newton published an explicit derivation of this relationship as a corollary of the law of universal gravitation in *Philosophiæ Naturalis Principia Mathematica* (*The Mathematical Principles of Natural Philosophy*). This derivation is widely available. Newton's version of the Law of Harmonies is as follows:

$$P^2 = \frac{4\pi^2}{G(M+m)}a^3,$$

where P is the orbital period of the planet in seconds, M is the mass of the star in kilograms, m is the mass of the planet in kilograms, a is the semi-major axis of the planet's orbit in meters, and G is the gravitational constant, which is approximately $6.674 \times 10^{-11} \text{ m}^3 \text{ kg}^{-1} \text{ s}^{-2}$. We can make this relationship linear by taking the natural log of both sides of the equation,

$$\begin{aligned}\ln(P^2) &= \ln\left(\frac{4\pi^2}{G(M+m)}a^3\right) \\ 2\ln P &= 3\ln a - \ln(M+m) + \ln\left(\frac{4\pi^2}{G}\right) \\ \ln P &= \frac{3}{2}\ln a - \frac{1}{2}\ln(M+m) + \frac{1}{2}\ln\left(\frac{4\pi^2}{G}\right).\end{aligned}\tag{1}$$

This linear equation now lends itself to a linear modeling approach. We will explore the extent to which the Law of Harmonies actually describes the orbital periods of planets observed in reality, and we will seek to discover what drives deviations from the law. To do this, we will be using data from the NASA Exoplanet Archive. Specifically, we will use the Planetary Systems Composite Data table, which can be found at the Nasa Exoplanet Archive [1].

This dataset contains data on all known exoplanets and their host star systems— an exoplanet is simply a planet orbiting a star other than our own Sun. The observational unit of this dataset is a planet. Each observation contains features related to the physical nature of the planet itself, the physical nature of the star(s) it orbits, the nature of the solar system it is in, the method by which it was discovered, and various other features like references to the scientific paper(s) in which the planet was first described, ...etc. The full complement of features are too many to enumerate here, and most of them are not of interest to us. The full list of features in the data can be found in the Data Column Definitions [2] section of the documentation page for the Planetary Systems Composite Data table. In the order given in the dataset, the features that are of interest to us in this analysis are as follows:

- `pl_name`: the name of the planet. This is simply a unique identifier for each observation.
- `sy_snum`: the number of stars in the system. This is a feature of the planet's system. Although rare, some solar systems have more than one star, e.g. the Alpha Centauri system, the closest known star system to our own, has three stars and two confirmed planets.
- `sy_pnum`: the number of planets in the system this planet inhabits.
- `cb_flag`: this is a binary flag denoting whether a planet is circumbinary. This is also rare, but some planet's are known to orbit two stars, e.g. the aptly named Kepler-16 system contains the planet Kepler-16b, which orbits two stars.
- `discoverymethod`: the method by which the planet was discovered. There are many ways to detect and gather information about a planet. We will discuss this later as a categorical variable, but the various methods of planet discovery are beyond the scope of this paper.
- `pl_orbper`: the orbital period of the planet given in Earth days.
- `pl_orbsmax`: the length of the semi-major axis of the planet's orbit given in astronomical units (AU). The path of a planet's orbit around its sun is an ellipse— this is Kepler's 1st Law. The semi-major axis of an ellipse is half of the largest possible diameter of the ellipse.
- `pl_rad`: the radius of the planet given in units of Earth Radii.
- `pl_bmasse`: the mass of the planet given in units of Earth Masses.
- `st_teff`: the stellar effective temperature given in degrees Kelvin. This would be the surface temperature at any point on the star if it was a black body, a perfect emitter of radiation. Oversimplified, this is the average surface temperature of the star.
- `st_rad`: radius of the star in units of Sun Radii.
- `st_mass`: mass of the star in units of Sun Masses.
- `st_lum`: luminosity of the star in units of Solar Luminosities.
- `sy_dist`: distance from Earth to the system in units of parsecs

Objectives

We wish to investigate the extent to which the Law of Harmonies actually describes planetary orbits observed in the universe. From taking the log of Newton's derivation, we have a specific, linear equation relating variables in our dataset. Then our first question follows naturally. Can we retrieve Newton's formulation of the Law of Harmonies by fitting a linear model of the same form?

Specifically, the final equation in (1) lends itself to the following linear model:

$$\ln P = \beta_0 + \beta_1 \ln a + \beta_2 \ln(M + m) + \epsilon, \quad (2)$$

using the appropriate variables from our dataset: `pl_orbper`, `pl_orbsmax`, `st_mass`, and `pl_bmasse` respectively. Fitting this model on our data, and adjusting the units of each variable to match those of G , (1) tells us that we should expect to see the following values for the coefficients of (2),

- $\beta_0 = \ln\left(\frac{4\pi^2}{G}\right)$
- $\beta_1 = \frac{3}{2}$
- $\beta_2 = -\frac{1}{2}$.

So, our first objective is to fit this model in an attempt to retrieve these coefficients, and, in the event that we do not retrieve the coefficients, we wish to investigate why.

Realistically, the Law of Harmonies would not be a famous result in physics if it did not work. We expect this model to be a very good fit. However, we recognize that this law is given for an ideal two-body system. That is, the law is derived with the assumption that we are dealing with one planet orbiting one sun in isolation. No system is truly isolated, and many systems have more than two bodies. Hence, we do not expect our model to be perfect in its predictive power. This gives rise to our second question. What drives deviations from the Law of Harmonies in our observations?

To investigate this question, we will fit a second model, using the residuals of the first as the target variable. We will use this model to look into which variables and factors contribute to an observation having a high residual. We can make a few guesses, based off informed intuition, about which variables we may expect to be helpful predictors of the residuals. The variables `sy_snum`, `sy_pnum`, and `cb_flag` are expected to be helpful in predicting the residuals because they encode information about how much a solar system deviates from an ideal, two-body system. We also expect some of `discoverymethod`, `pl_rade`, `pl_bmasse`, `st_lum`, and `sy_dist` to be helpful predictors because they encode information related to how the planet is discovered and measured.

The data is very clean, so we do not expect any issues with the data itself. The only potential issue we might run in to in the modeling is multicollinearity in some of the covariates that describe the physical features of our objects. We will address this, if necessary, in our analysis.

Analysis

All code written for this analysis is written in R apart from some limited data preprocessing steps, which are done in Python via Pandas. Additionally, a significance level of 0.05 is used to test any p-values in our analysis.

Data Preprocessing

This dataset is very clean and tidy as it is. Hence, the data preprocessing is minimal. First, we drop all columns from the data apart from those listed in the introduction. Then, all variables given in non-metric units are converted to the appropriate base unit in the metric system. Lastly, we ensure that the data is properly typed. For this analysis, we will treat the variables `sy_snum`, `sy_pnum`, `cb_flag`, and `discoverymethod` as categorical, so they are converted to factors in R. The variable `pl_name` is also treated as categorical, but it is only an identifier so we leave it as the type character in R. All other variables are treated as numerical, and they are converted to that type in R.

Exploratory Data Analysis

Before diving in to our full analysis, we do some exploratory data analysis (EDA) to get a feel for our data, and ensure that we are not including any redundant variables. The first thing we wish to see is the pairwise correlations between all numeric variables in our data.

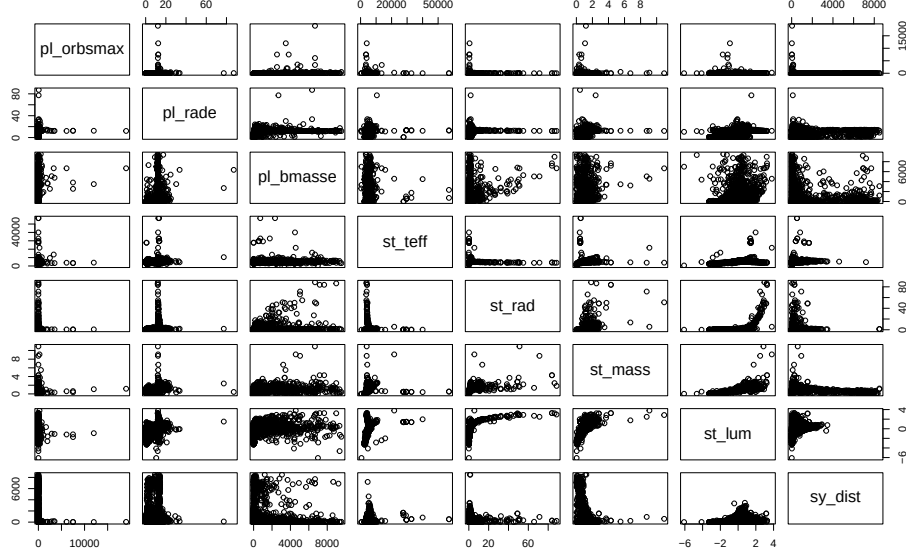


Figure 1: Pairs Plot of Numeric Variables

For purposes of readability, we have also dropped our target variable, `pl_orbper`, from Figure 1. We see directly from Figure 1 that we don't have any worrying correlations other than that between `st_mass` and `st_lum`. The correlation coefficient between these two is 0.78, so we will exclude `st_lum` from our analysis henceforth. We anticipated that `st_lum` may be helpful in our analysis of the residuals because the brightness of a star effects the measurements made when a planet is in transit. While in transit the planet exerts an effect on the light of the star, from which information about the planet can be inferred. The brighter the star, the easier these affects are to measure. However, we see that it is generally true that more massive stars are brighter, so we can use stellar mass as a proxy for this effect. It should be noted that we are comfortable making this assumption because of the lack of correlation between `st_mass` and `st_rad`—larger stars, spatially, can obscure transit measurements. Moreover, we wish to capture the effects of stellar mass in our model, so we make the decision to drop `st_lum` rather than `st_mass`.

Retrieving the Law of Harmonies

Now we move on to fitting the model outlined in (2).


```

Call:
lm(formula = log(pl_orbper) ~ log(pl_orbsmax) + log(st_mass +
  pl_bmasse), data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-1.9509 -0.0142 -0.0024  0.0132  3.4468

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    11.8952393   0.2148778   55.36  <2e-16 ***
log(pl_orbsmax)  1.4973584   0.0009227 1622.86  <2e-16 ***
log(st_mass + pl_bmasse) -0.4752727   0.0031424 -151.24  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09607 on 5491 degrees of freedom
(784 observations deleted due to missingness)
Multiple R-squared:  0.998, Adjusted R-squared:  0.998
F-statistic: 1.343e+06 on 2 and 5491 DF, p-value: < 2.2e-16

```

The first thing to notice is that this is a very good model. We have an R-squared value of 0.996 It doesn't get much better than that. Secondly, we look at the coefficients. The coefficient for $\log(\text{pl_orbsmax})$ is 1.497. This is exactly what we expected from the theory, with $\beta_1 = \frac{3}{2}$. The coefficient of $\log(\text{st_mass} + \text{st_bmasse})$ is -0.475 . This also aligns with our predicted value of $\beta_2 = -\frac{1}{2}$. However, our intercept is 11.895. We predicted this value as $\beta_0 = \ln\left(\frac{4\pi^2}{G}\right)$, which evaluates to 13.553. So, with this simplest model, we were able to retrieve the slope coefficients from Newton's derivation, but we have a discrepancy in our intercept.

Seeing that we have a discrepancy in our intercept immediately makes us suspicious that we may have some high leverage values in our data that are pulling the line up or down. To get an idea of this, we can take a look at the graph of fitted values vs residuals for our model.

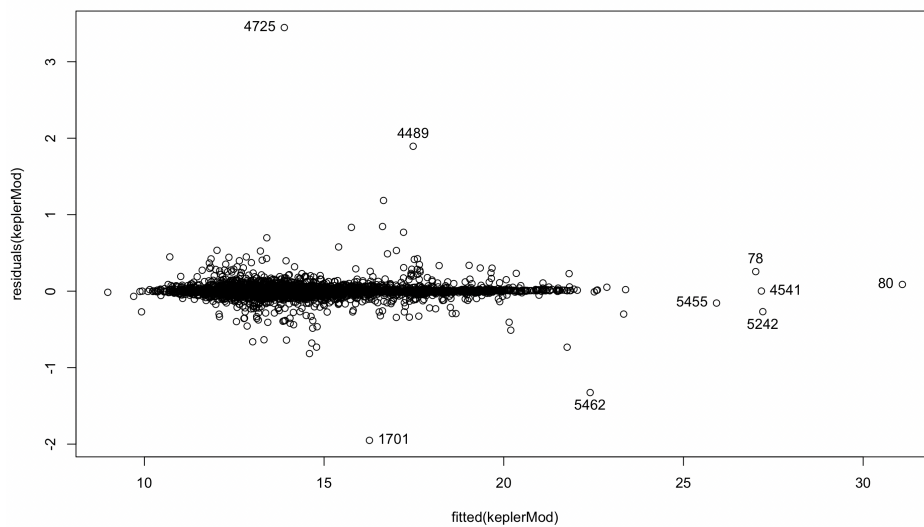


Figure 2: Fitted vs Residual Values for (2)

Figure 2 gives us some candidates for which observations may be altering our intercept. Namely, they are observations 5455, 5242, 4541, 78, and 80. We can also benefit from looking at a graph of residuals vs leverage for this model.

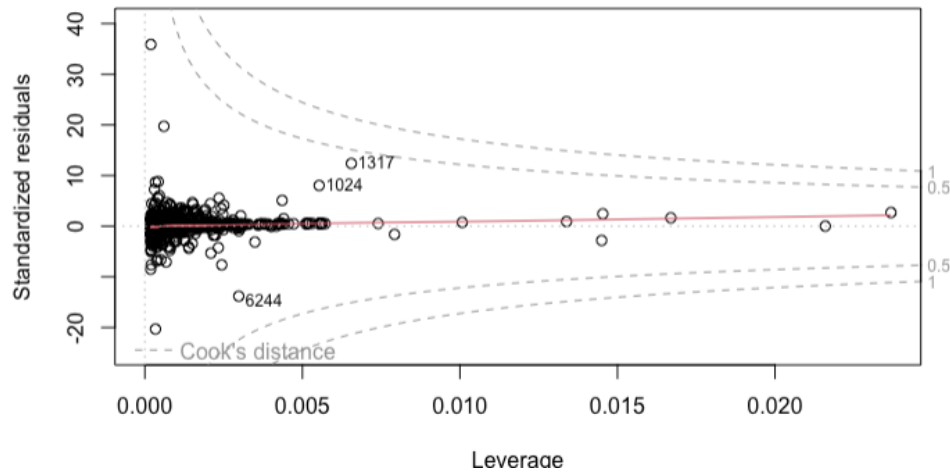


Figure 3: Residual Values vs Leverage for (2)

This graph gives us three more candidates for values that may be causing the discrepancy in our intercept– 1317, 1024, and 6244.

To check the effect that these entries have on our coefficients we can take a look at the DFBETA values corresponding to each of these observations.

obs_id	(Intercept)	log(pl_orbsmax)	log(st_mass + pl_bmasse)
78	0.0012774554	0.0017639907	-0.0014197776
80	0.0004482796	0.0009194878	-0.0005277042
1024	-0.5739196162	0.0675354506	0.5573203331
1317	-0.9882380852	0.0472850199	0.9664458204
4541	0.0009773380	0.0007332136	-0.0010408617
5242	0.0002177052	0.0001525774	-0.0002327953
5455	0.0095367811	-0.0018300754	-0.0091565057
6244	-0.1692306881	-0.7417971261	0.2391697539

To help us judge the DFBETA values, we look at $\frac{2}{\sqrt{n}} = 0.027$, where n is the number of observations used in fitting the model. With this value in mind, we can see that the observations with high Cook's distances are the ones with large DFBETA values for every parameter. Now we will see how removing these values changes the coefficients of our model.

```
Call:
lm(formula = log(pl_orbper) ~ log(pl_orbsmax) + log(st_mass +
  pl_bmasse), data = df_clean, na.action = na.exclude)

Residuals:
    Min       1Q   Median       3Q      Max
-1.9513 -0.0123 -0.0021  0.0121  3.4468

Coefficients:
              Estimate Std. Error t value Pr(>|t|)
(Intercept)    12.2648766   0.2079364   58.98  <2e-16 ***
log(pl_orbsmax)  1.4979251   0.0008891 1684.84  <2e-16 ***
log(st_mass + pl_bmasse) -0.4807742   0.0030403  -158.13  <2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09244 on 5488 degrees of freedom
(784 observations deleted due to missingness)
Multiple R-squared:  0.9981,    Adjusted R-squared:  0.9981
F-statistic: 1.447e+06 on 2 and 5488 DF,  p-value: < 2.2e-16
```

So we see that removing these points actually brought all three coefficients closer to their theoretically predicted values. However, upon analysis of the data points in question, we can find no reason to justify removing them from the data.

Modeling Residuals

Now we move on to addressing our second question. We will use the residuals of our first model as the target variable for our second model. Before we move to model selection, we take a look at the boxplots of the residuals split by the four factors in our data.

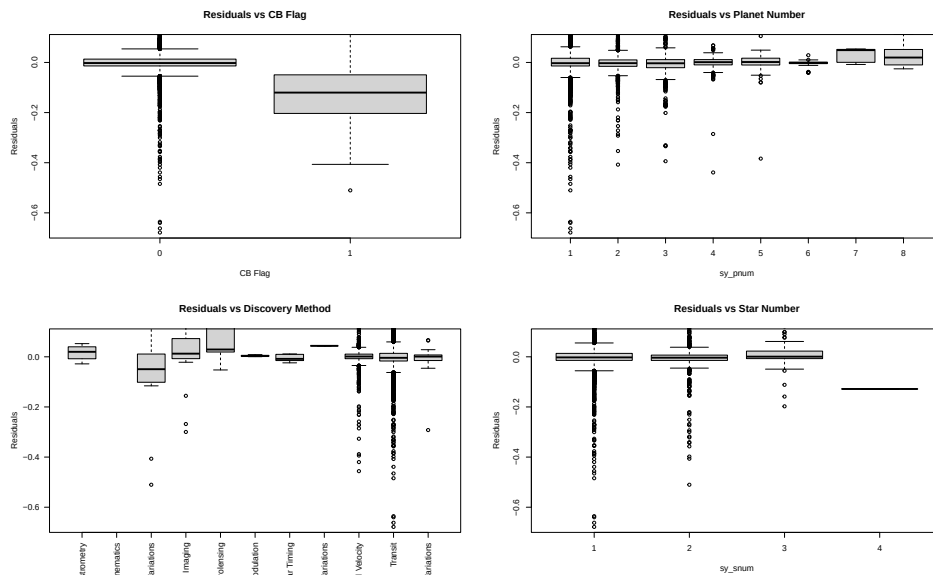


Figure 4: Boxplots of Residuals per Various Factors

Figure 4 tells us that we can expect to see significant effects in an ANOVA model between the levels in `cb_flag` and in `discoverymethod`.

We begin our model selection by fitting a full model of all the factors in our data, including interaction terms. We then use backward elimination according to the following heuristics. We perform a simple ANOVA test on the model. We select the variable with the largest p-value and fit a new model with this variable removed. Then we perform an ANOVA test between the two models. If the model is improved by removing the variable, we keep the simpler model. Then, we repeat this process with the new model. This is repeated until the model is no longer improved by removing a variable. The output of the steps in this process is not included in this report, but we do include a summary of our final model. The model ended up only including terms `cb_flag` and `discoverymethod`, with no interactions.

```
Call:
lm(formula = residuals ~ (cb_flag + discoverymethod), data = df)

Residuals:
    Min       1Q   Median       3Q      Max
-1.9489 -0.0150 -0.0032  0.0130  3.4386

Coefficients:
                                Estimate Std. Error t value
(Intercept)                   0.016057   0.047369   0.339
cb_flag1                      -0.188223   0.020434  -9.211
discoverymethodEclipse Timing Variations  0.113270   0.057467   1.971
discoverymethodImaging          0.030777   0.051077   0.603
discoverymethodMicrolensing      0.213458   0.054724   3.901
discoverymethodOrbital Brightness Modulation -0.011988   0.072358  -0.166
discoverymethodPulsar Timing     -0.021789   0.061154  -0.356
discoverymethodPulsation Timing Variations  0.027635   0.105921   0.261
discoverymethodRadial Velocity   -0.007849   0.047452  -0.165
discoverymethodTransit          -0.018017   0.047392  -0.380
discoverymethodTransit Timing Variations -0.018503   0.050078  -0.369
Pr(>|t|)
(Intercept)                   0.7346
cb_flag1                      < 2e-16 ***
discoverymethodEclipse Timing Variations  0.0488 *
discoverymethodImaging          0.5468
discoverymethodMicrolensing      9.71e-05 ***
discoverymethodOrbital Brightness Modulation 0.8684
discoverymethodPulsar Timing     0.7216
discoverymethodPulsation Timing Variations 0.7942
discoverymethodRadial Velocity   0.8686
discoverymethodTransit          0.7038
discoverymethodTransit Timing Variations 0.7118
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Residual standard error: 0.09474 on 5483 degrees of freedom
(784 observations deleted due to missingness)
Multiple R-squared:  0.02901,    Adjusted R-squared:  0.02724
F-statistic: 16.38 on 10 and 5483 DF,  p-value: < 2.2e-16
```

The summary of our model tells us that we do have a significant effect from the `cb_flag` levels and from `discoverymethodMicrolensing`. To further investigate these effects, we can perform Tukey's Honest Significant Difference Test on the two factors. For `cb_flag`, we see the following,

```

$statistics
      MSerror   Df      Mean      CV
0.008975456 5483 3.501022e-18 2.706035e+18

$parameters
      test name.t ntr StudentizedRange alpha
Tukey cb_flag  2      2.77242  0.05

$means
      residuals      std      r      se      Min      Max      Q25
0  0.0008409775 0.09492727 5458 0.001282365 -1.9508633 3.4468024 -0.01391645
1 -0.1275015397 0.16209159  36 0.015789814 -0.5103274 0.3475076 -0.17947741
      Q50      Q75
0 -0.002338603 0.01329253
1 -0.120272502 -0.05005100

$comparison
      difference pvalue signif.      LCL      UCL
0 - 1  0.1283425      0      *** 0.0972863 0.1593987

$groups

```

We knew what the results would be from this test both from the theory, and from the fact that it was significant in our ANOVA model and it only has two levels. Still, we see that there is a statistically significant difference in the mean residual associated with circumbinary star vs normal stars, with the model for the orbital period consistently overestimating for circumbinary stars. We do not include all the output of the HSD test for discovery method, as it is too verbose. We only include the means and the comparisons.

```

$means
      residuals      std      r      se
Astrometry      0.016057225 0.033795620  4 0.047369441
Eclipse Timing Variations -0.058896554 0.209061060 14 0.025320031
Imaging      0.024247730 0.129549920 25 0.018947776
Microlensing      0.213829861 0.537345909 12 0.027348760
Orbital Brightness Modulation 0.004069091 0.004197801  3 0.054697519
Pulsar Timing      -0.005732005 0.014146457  6 0.038676987
Pulsation Timing Variations 0.043692091      NA  1 0.094738882
Radial Velocity      0.007549261 0.135120210 1142 0.002803465
Transit      -0.002579691 0.076469756 4253 0.001452715
Transit Timing Variations -0.002446050 0.061925056  34 0.016247584

```

Astrometry - Eclipse Timing Variations	0.9290	
Astrometry - Imaging	1.0000	
Astrometry - Microlensing	0.0113	*
Astrometry - Orbital Brightness Modulation	1.0000	
Astrometry - Pulsar Timing	1.0000	
Astrometry - Pulsation Timing Variations	1.0000	
Astrometry - Radial Velocity	1.0000	
Astrometry - Transit	1.0000	
Astrometry - Transit Timing Variations	1.0000	
Eclipse Timing Variations - Imaging	0.2042	
Eclipse Timing Variations - Microlensing	0.0000	***
Eclipse Timing Variations - Orbital Brightness Modulation	0.9895	
Eclipse Timing Variations - Pulsar Timing	0.9794	
Eclipse Timing Variations - Pulsation Timing Variations	0.9894	
Eclipse Timing Variations - Radial Velocity	0.2137	
Eclipse Timing Variations - Transit	0.4426	
Eclipse Timing Variations - Transit Timing Variations	0.6852	
Imaging - Microlensing	0.0000	***
Imaging - Orbital Brightness Modulation	1.0000	
Imaging - Pulsar Timing	0.9995	
Imaging - Pulsation Timing Variations	1.0000	
Imaging - Radial Velocity	0.9973	
Imaging - Transit	0.9239	
Imaging - Transit Timing Variations	0.9875	
Microlensing - Orbital Brightness Modulation	0.0215	*
Microlensing - Pulsar Timing	0.0002	***
Microlensing - Pulsation Timing Variations	0.7811	
Microlensing - Radial Velocity	0.0000	***
Microlensing - Transit	0.0000	***
Microlensing - Transit Timing Variations	0.0000	***
Orbital Brightness Modulation - Pulsar Timing	1.0000	
Orbital Brightness Modulation - Pulsation Timing Variations	1.0000	
Orbital Brightness Modulation - Radial Velocity	1.0000	
Orbital Brightness Modulation - Transit	1.0000	
Orbital Brightness Modulation - Transit Timing Variations	1.0000	
Pulsar Timing - Pulsation Timing Variations	1.0000	
Pulsar Timing - Radial Velocity	1.0000	
Pulsar Timing - Transit	1.0000	
Pulsar Timing - Transit Timing Variations	1.0000	
Pulsation Timing Variations - Radial Velocity	1.0000	
Pulsation Timing Variations - Transit	1.0000	
Pulsation Timing Variations - Transit Timing Variations	1.0000	
Radial Velocity - Transit	0.0439	*
Radial Velocity - Transit Timing Variations	0.9999	
Transit - Transit Timing Variations	1.0000	

Here we see that planets discovered via microlensing appear to behave differently than planets discovered via all other methods of discovery, with the original model consistently underestimating the correct orbital period.

It may also be valuable to take a non ANOVA approach to modeling the residuals. We can fit a simple, additive model with residuals as the target and the variables `pl_rade`, `st_teff`, `st_rad`, `st_mass`, and `sy_dist` as input. Recall that we decided to drop stellar luminosity. We then reduce the model by removing those predictors with insignificant p-values. The result is the removal of `pl_rade` from the model. However, while the model coefficients are statistically significant, they have coefficients so close to zero as to be negligible.

Conclusions

The main finding of our analysis is that the Law of Harmonies is very resilient to deviations from ideal, two-body systems. This is not entirely unexpected, it is a famous result after all, but the extent to which it is resilient is surprising. In fact, the only time that we can detect that the predictive power of the law starts to break down is in circumbinary star systems. Again, this is expected. We are not surprised to see the effect of circumbinary star systems, but we are surprised that we did not detect an effect from star systems with variant numbers of stars. One might suspect that the gravitational influence of an extra star in a solar system might cause orbits to deviate from the law, but we found no significant difference between the number of stars in a system with respect to the residuals of the model for predicting orbital period.

The deviations among residual means for the method of planet discovery are interesting, particularly in the case of discovery by microlensing. It is tempting to conclude that microlensing truly behaves differently from other methods, but our model only included 12 planets found via microlensing. Considering the size of the dataset, a confident conclusion cannot be made on this matter without more examples of planets found by microlensing. Of course, this dataset contains every known exoplanet, so there is no more data on the topic. Hopefully, this will change as time goes on and analyzing residual error associated with microlensing observations could be a compelling area of future research.

It is also worth mentioning the fact that we did not detect any relationship between the variables `pl_rade`, `st_teff`, `st_rad`, `st_mass`, and `sy_dist` and the residuals of our model. We expected stellar mass, stellar radius, and planetary radius to have effects on things like transit measurements, contributing to residuals. We did not find this to be the case. It is also intuitive to think that the farther away a system is, the less accurate the measurements on that system will be, contributing to higher residuals. Again, we did not find this to be the case. It may be fruitful in the future to consider an interaction model with `discoverymethod` and some of these variables to further investigate these relationships.

The strength of this analysis is also the weakness of this analysis. The Law of Harmonies works—very well. This made fitting our initial model very effective, but it came with the drawback that the residuals were difficult to analyze. In the future, it may be useful to include variables like orbital eccentricity or other variables that describe the geometry of a planets orbit, e.g. orbital inclination.

Works Cited

- [1] Planetary systems composite data, <https://exoplanetarchive.ipac.caltech.edu/cgi-bin/TblView/nph-tblView?app=ExoTbIs&config=PSCompPars> (accessed May 7, 2026).
- [2] “Planetary systems and planetary systems composite parameters table definitions,” Planetary Systems and Planetary Systems Composite Parameters Data Column Definitions, https://exoplanetarchive.ipac.caltech.edu/docs/API_PS_columns.html (accessed May 7, 2026).